



Commentary

Climate change is a threat multiplier for violence against children[☆]

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ABSTRACT

Background: The climate crisis is the biggest threat to the health, development, and wellbeing of the current and future generations. While there is extensive evidence on the direct impacts of climate change on human livelihood, there is little evidence on how children and young people are affected, and even less discussion and evidence on how the climate crisis could affect violence against children.

Participants and setting: In this commentary, we review selected research to assess the links between the climate crisis and violence against children.

Methods: We employ a social-ecological perspective as an overarching framework to organize findings from the literature and call attention to increased violence against children as a specific, yet under-examined, direct and indirect consequence of the climate crisis.

Results: Using such a perspective, we examine how the climate crisis exacerbates the risk of violence against children at the continually intersecting and interacting levels of society, community, family, and the individual levels. We propose increased risk of armed conflict, forced displacement, poverty, income inequality, disruptions in critical health and social services, and mental health problems as key mechanisms linking the climate crisis and heightened risk of violence against children. Furthermore, we posit that the climate crisis serves as a threat multiplier, compounding existing vulnerabilities and inequities within populations and having harsher

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consequences in settings, communities, households, and for children already experiencing adversities.

Conclusions: We conclude with a call for urgent efforts from researchers, practitioners, and policymakers to further investigate the specific empirical links between the climate crisis and violence against children and to design, test, implement, fund, and scale evidence-based, rights-based, and child friendly prevention, support, and response strategies to address violence against children.

1. Introduction

Human activities and limited action by governments, corporations, and global organizations continue to lead to unprecedented levels of environmental stress (Pörtner et al., 2022). Without more comprehensive changes in policy, practice, and accountability, by the end of the century the planet is on track to heat up by between 2.1 and 2.9 degrees compared to pre-industrial levels, which could undermine the viability of human societies (Ripple et al., 2022). In this paper, we employ a social-ecological perspective to call attention to increased violence against children as a specific, yet under-examined, direct and indirect consequence of the climate crisis.

Ecological degradation and the related outcomes of the climate crisis have put human natural systems under unprecedented levels of strain. Global emissions hit a record high in 2022 (Tollefson, 2022). Warming-induced extreme weather events, such as drought, wildfires and flooding, have increased in number and intensity (World Meteorological Association, 2022). Degraded air and water quality, resource scarcity (e.g., fertile land), rising sea levels, and increased spread and transmission of infectious and zoonotic diseases are causing widespread suffering and loss of life (Kemp et al., 2022; Romanello et al., 2021; World Health Organization, 2021).

Currently, nearly half the world's children live in countries at extremely high risk of ongoing and worsening exposure to climate and environmental shocks (Unicef, 2021). As a result, the climate crisis will inflict compounding and unequal harms on progressive generations of children, devastating their health and wellbeing, livelihoods, and cultures. Evidence from humanitarian crises or emergencies, including the COVID-19 pandemic, suggest that children are among the most impacted (Bhatia et al., 2021; Bourgault, Peterman, & O'Donnell, 2021; McCoy et al., 2021). Children are growing up in a context where inequality within and across nation-

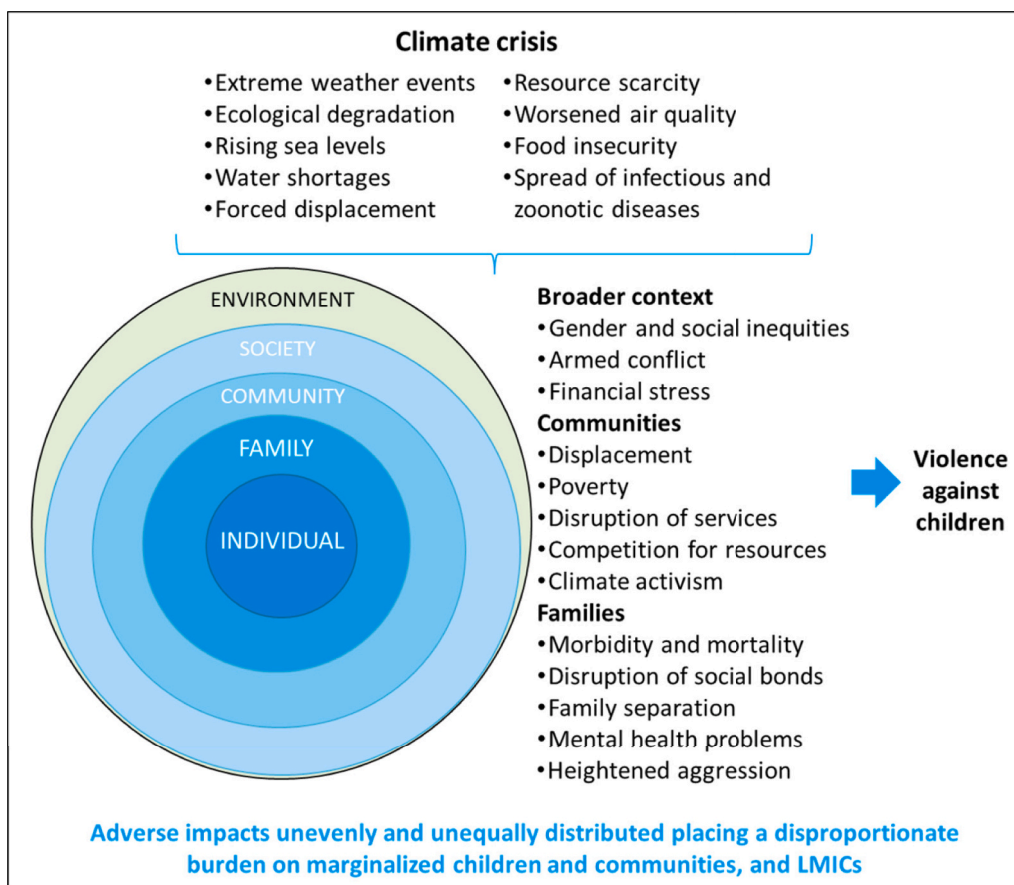


Fig. 1. A social ecological perspective on the direct and indirect pathways linking the climate crisis and violence against children.

states has deepened due to intensifying environmental emergencies and overlapping global crises such as the COVID-19 pandemic and multiple acute and protracted humanitarian crises (Chancel, Piketty, Saez, & Zucman, 2022). These forms of structural injustices, including the misuse and abuse of environmental resources and ecosystems, are sometimes conceptualized as forms of structural violence (Farmer, 2009; Galtung, 1969). Nixon (2011) calls this *slow violence*, an attritional violence whose calamitous repercussions are postponed for years or decades or centuries. Therefore, the climate crisis is a child rights and a child health issue.

Although the impacts of the climate crisis – as a form of structural violence with intergenerational consequences – on the health and wellbeing of adults have been well documented (e.g., Romanello et al., 2022), there is more limited evidence on how children and young people are affected, and even less evidence on how the climate crisis could affect violence against children. We conceptualize violence against children as the use of physical force or power, threatened or actual, that results in, or has a likelihood of resulting in, harm, death, psychological injury, maldevelopment or deprivation that can occur in different settings like the home, school, and community (World Health Organization, 2016). Violence can be physical, psychological, and sexual, as well as abandonment, neglect, forced marriage, witnessing violence, and child labor. In this commentary, we clarify conceptual links between the climate crisis and a heightened risk of interpersonal violence against children, drawing on the current evidence and from lessons from the COVID-19 pandemic and other crises. We conclude the commentary with a call for urgent efforts from researchers, practitioners, and policy-makers to further investigate the specific empirical links between the climate crisis and violence against children and to design, test, implement, fund, and scale evidence-based, rights-based, and child friendly prevention, support, and response strategies to address violence against children.

2. A social-ecological perspective on the climate crisis and violence against children

While the specific consequences of the environmental crisis are complex and context specific, climate-related disasters and protracted environmental crises lead to cascading negative impacts on social and economic outcomes (Baxter et al., 2022; Romanello et al., 2021), which will be experienced unequally and unevenly by children and young people with greater impacts for marginalized children.

A social-ecological perspective (Bronfenbrenner & Morris, 2007; Cicchetti, Toth, & Maughan, 2000; Sameroff, 2010) is a useful frame for examining how the climate crisis exacerbates the risk of violence against children at the continually intersecting and interacting levels of society, community, family, and the individual (see Fig. 1; Cerna-Turoff et al., 2021; Maternowska & Fry, 2018; Ward, Grogan-Kaylor, Pace, Cuartas, & Lee, 2021). At the societal and community level, the climate crisis can exacerbate the risk of armed conflict, political and targeted violence, and forced displacement, increase contextual stressors like poverty and income inequality, and disrupt health and social services and supports to families. At the family and individual levels, it can also undermine family and community bonds, trigger caregivers' morbidity and mortality, and negatively affect both caregivers' and children's mental health. In what follows, we discuss pathways and mechanisms at each level of the social ecology. Importantly, many of the mechanisms we highlight may also lead to increased intimate partner violence and other forms of gender-based and contextual violence, which intersect with violence against children across the life course and further intensify the risk of negative outcomes for children, women, families, and communities.

At the society and community levels, extreme climate events can multiply the risk of armed conflict and political violence, especially in regions more dependent on agriculture or with higher pre-existing fragility and political marginalization (Koubi, 2019; Levy, Sidel, & Patz, 2017; Mach et al., 2019). While the relationship between climate and conflict is complex and context-specific, increased food insecurity, water shortages and competition for other increasingly scarce resources have been identified as exacerbating factors in a number of ongoing conflicts (Breckner & Sunde, 2019). Prior research has identified associations between extreme temperatures (Bollfrass & Shaver, 2015; Maystadt, Calderone, & You, 2014), droughts (von Uexkull, 2014), floods (Ghimire & Ferreira, 2016), and atypical rainfall patterns (Salehyan & Hendrix, 2014) with armed conflict across regions and countries. In turn, armed conflict can lead to child victimization, including death, forced recruitment of children into armed groups, and physical injury, while also increasing the risk of other forms of violence like abandonment (due to the death of adult caregivers), neglect, forced marriage, trafficking, sexual abuse, and witnessing contextual violence and other traumatic events (Seddighi, Salmani, Javadi, & Seddighi, 2019; Shenoda et al., 2018).

More frequent and/or severe weather events, sea-level rise, and the effects of the climate crisis on food, soil, and water availability can lead to increased forced displacement and may increase humanitarian crises. Approximately 89.3 million people worldwide were forcibly displaced at the end of 2021 for several reasons, including conflict and climate-related risks (UNHCR, 2021). Yet, it is estimated that double this number of people are at risk of forced displacement because of sea-level rise alone, as well as more than 50 million people are at risk due to drought, floods, and other extreme weather events. By 2030, about 700 million people in Africa alone may be displaced by drought (Kam et al., 2021; Levy et al., 2017; Pörtner et al., 2022; Traore Chazalnoël, Ionesco, & Duca, 2020). In turn, forced displacement can trigger and/or intensify vulnerabilities and risks for violence against children due to disruptions in family and community bonds, lack of access to basic social services and sources of protection (like schools), and contextual threats that might increase the risk of child labor, transactional sex, forced recruitment into armed groups, orphaning, neglect, and sexual, psychological, and physical violence directed at children or their main caregivers (Mels, Derluyn, Broekaert, & Rosseel, 2010; UNHCR, 2021).

Poverty and resource scarcity caused by the climate disaster can also intensify the risk of violence against children through other mechanisms. Climate-related resource scarcity, financial stress, and food insecurity can increase stress and aggression among males, women's disempowerment, and intra-household inequalities, intensifying child and gender-based violence (Anastario, Shehab, & Lawry, 2009; Epstein, Bendavid, Nash, Charlebois, & Weiser, 2020; Rezwana & Pain, 2021). Food insecurity can also increase child

labor, recruitment by armed groups, forced child marriage, and risk of sexual trafficking as families seek ways to meet their basic needs (Plan International, 2022). Even when families are not displaced or directly exposed to violence, prior evidence shows that children and women tend to be forced to walk farther and stay for longer in places far from their communities in search of food, water, and other resources when hit by droughts, floods, and other climate disasters, increasing the risk of sexual and physical violence (Tallman et al., 2023).

Within their communities and societies, children not only experience climate harms, but they also increasingly participate in efforts to address this crisis (Neas, Ward, & Bowman, 2022; Ritchie, 2021). The risks of violence facing children engaged in local struggles for land and natural resources is a critical and under-examined topic. Efforts for climate adaptation and mitigation at the community level may not always include the needs of children. Evidence shows environmental defenders worldwide experience increasing levels of criminalization, threats, intimidation, and physical violence from both state and private actors (Grant & Le Billon, 2020). Yet children appear in this work only anecdotally, if at all. Meanwhile, it is clear that children can and do face real and threatened violence because of their activism (Daniels, 2022; United Nations Committee on the Rights of the Child, 2018).

At the family and individual levels, mental health and caregiver stress is a key mechanism linking the climate crisis and violence against children. Recent reviews of the literature indicate that both acute (e.g., floods, wildfires) and chronic (e.g., temperature change) climate-related events can lead to mental health problems like anxiety, mood disorders, stress, sleep disorders, aggressive behavior, and suicidal ideation (Cianconi, Betrò, & Janiri, 2020; Cuijpers et al., 2023; Palinkas & Wong, 2020). For example, one study using retrospective data from the U.S. estimated that a 1 °C five-year warming might lead to a 2 % increase in the prevalence of mental health problems (Obradovich, Migliorini, Paulus, & Rahwan, 2018); yet, this figure might be an underestimation of the actual problem given other risks to mental health triggered by the climate crisis. Other studies estimated that warmer temperatures might lead to linear increases in interpersonal violence (Hsiang, Burke, & Miguel, 2013; Miles-Novelo & Anderson, 2019). While the climate crisis might exacerbate mental health challenges through trauma, financial stress, impacts on physical health (e.g., heat exposure causes heat exhaustion), and impacts on communities (Berry, Bowen, & Kjellstrom, 2010), a survey with 10,000 children and young people in ten countries also show widespread anticipatory climate anxiety, including sadness, anger, powerlessness, helplessness, and guilt (Hickman et al., 2021).

The compounding impacts of the climate crisis, which puts unbearable strain on families, might deplete cognitive and emotional resources that caregivers require to protect their children and respond to their emotional and physical needs, which in turn increases the risk of neglect and makes it more challenging for caregivers to identify and respond to other contextual threats to children's safety. According to cognitive and behavioral science, these psychosocial impacts can also make it more difficult for parents to inhibit impulsive and aggressive behaviors (e.g., physical punishment) in the face of daily stressors, heightening the risk for physical and psychological violence (Cerna-Turoff et al., 2021; Holden, 2020; Sanson, Malca, & Van Hoorn, 2022).

In addition to direct mechanisms, disruptions in formal and informal social safety nets and social programs and services (e.g., physical and mental health services, social assistance programs, parenting programs) can further intensify the risk of violence against children. Theory and evidence associate limited caregivers' knowledge on early childhood development and parenting, inadequate expectations on normative child development, and positive attitudes toward authoritarian parenting as a key risk factor for physical punishment and other forms of violence against children (Cerna-Turoff et al., 2021; Holden, 2020; Ward et al., 2021). Over the past years, several services in the education, health, and social protection systems, like nutrition and parenting programs, have intended to address these “cognitive” risk factors, demonstrating positive impacts on reducing violence against children (Knerr, Gardner, & Cluver, 2013; Leijten et al., 2019). Yet, the direct consequences of the climate crisis may lead to overwhelmed and/or disrupted social services (Baxter et al., 2022; Romanello et al., 2021), making it more difficult for families to access basic information and support on child development and childrearing and to engage in positive parenting behaviors.

Furthermore, the climate crisis serves as a threat multiplier, compounding existing vulnerabilities and inequities within populations and having harsher consequences in settings, communities, households, and for children already experiencing adversity (Baxter et al., 2022; Pelling & Garschagen, 2019). Put differently, the pathways and mechanisms we explore above will not affect children equally. At the societal and community levels, the poorest countries have experienced (and will likely continue experiencing) the worst consequences from the climate crisis, even though their contribution to the crisis itself is insignificant relative to wealthiest countries (Magnan et al., 2021; Romanello et al., 2021). Within countries, prior climate disasters indicate that the poorest households are more likely to be affected by extreme weather events and temperature change relative to wealthiest households (Hallegatte & Rozenberg, 2017; Pelling & Garschagen, 2019). In addition to the unequal distribution of risks and consequences, intersectional and cumulative risk perspectives suggests that racially and/or ethnically marginalized groups and communities (e.g., Indigenous groups), families experiencing poverty, pre-existing mental health problems, having children with disabilities, or facing adversities more generally might be more fragile and vulnerable to new stressors (Center on the Developing Child at Harvard University, 2016; Walker et al., 2011). Therefore, low-income countries, families, and groups experiencing discrimination, oppression, marginalization, and adversity may face particularly strong difficulties responding and adapting to climate-related threats, further intensifying the risk of violence against children.

3. Next steps for research and action

In line with the statement by the Special Representative of the Secretary General on Violence Against Children at the General Assembly in July 2022, existing research makes it clear that the climate crisis already inflicts direct and structural violence on the children of today and tomorrow. A growing evidence base shows that the harms caused by the crisis are unequally distributed, hitting especially hard children and families who are already at risk (Unicef, 2021). More ambitious mitigation actions are urgently needed,

particularly by the “minority world” (also known as wealthy/industrialized or high-income countries; Khan, Abimbola, Kyobutungi, & Pai, 2022), recognizing that while these countries are making the largest contribution to the climate crisis, the “majority world” (i.e., low- and-middle income countries) tend to experience worse consequences. Yet, for children who are already suffering the impacts of the climate crisis there is also a pressing need for support, planning and action to promote communities' and families' adaptation and resilience to prevent the collateral consequences of the climate crisis, including violence against children. Such efforts should be child centered, rights based and meaningfully engage children in their design. There is an urgent need to make visible the links between climate impacts and violence against children, critically supported by more research exploring the specific consequences of climate change on violence against children, as well as the distribution of these effects across and within countries to ensure equity and social inclusion are at the center of policies and intervention strategies.

Regarding action, there is a need for innovative, low-cost, scalable, and culturally sensitive interventions and violence prevention strategies addressing risk factors and promoting sources of protection at each level of the social ecology (society, community, family, and individual levels). At the society and community levels, there is a need for recognition that the climate crisis and violence against children are cross-cutting issues that require national, regional, and multi-sectoral coordination and financing to ensure families' and children's access to prevention and response programs and justice. There is an urgent need for increased climate-responsible investments in preventative and protective programs and existing social protection structures, child protection measures within humanitarian settings, and systematic inclusion of children and child protection within all climate planning and policy, considering ecological, intersectional, and cumulative risk perspectives. There is also a need for more strategies aimed at changing social norms that legitimize and promote violence against children, using mass media and collaborating with political, religious, and social leaders.

At the community, family, and individual levels, there is a need for strong social protection systems, poverty alleviation, and efforts to improve food security and reduce caregiver stress, including culturally-relevant, scalable strategies aimed at promoting core life skills (e.g., emotional awareness, self-regulation, mental flexibility) for both adults and children to support adaptation in the face of future stressors and uncertainty. Moreover, considering that the climate disaster may impact children's development and behavior (Sansón et al., 2022), there is also a need to provide support for parents and caregivers as they navigate the climate crisis to understand and be responsive to their children's needs. Parenting and caregiver support programs are a promising strategy to promote social norm change, caregivers' knowledge of parenting and their children's needs, and parent and child mental health core life skills (Coore Desai, Reece, & Shakespeare-Pellington, 2017; Cuartas, Baker-Henningham, Cepeda, Rey-Guerra, & ICBF Team, 2022; Jeong, Franchett, Ramos de Oliveira, Rehmani, & Yousafzai, 2021; Knerr et al., 2013; Lachman et al., 2017). Yet, disruptions in social services and barriers for in-person interactions amid the COVID-19 pandemic, demonstrated that there is need for fresh-thinking and innovation in the way parenting programs are delivered to provide adequate and uninterrupted support at scale for families (Shenderovich et al., 2021). This includes moving away from proprietary programs toward more open-source, program-agnostic solutions that allow for cross-cultural adaptation and innovation across contexts.

We conclude with a call for structural and societal action. Besides urgent action to mitigate the climate crisis (i.e., radical decreases in the emissions of greenhouse gas into the atmosphere), there is a need to revise definitions, research agendas, policy actions, and legal frameworks so that societies recognize climate destruction as a (intentional or unintentional) trigger of violence against children and against future generations. Taking young people's participation and empowerment seriously in these tasks will be critical to ensure children's protection from violence and secure a sustainable future.

Data availability

No data was used for the research described in the article.

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